

Medical Robotics

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Lecture 11

Registration Case Study

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1. Introduction to Oncological Hyperthermia

While previous lectures focused on registration within orthopedic applications, this case study explores the registration problem for an experimental robot-assisted system designed for superficial hyperthermia. Although the involved reference frames and medical application are distinct, the underlying mathematical algorithms for registration remain fundamentally the same.

Oncological Hyperthermia (HT) is a therapeutic treatment for superficial or deep cancerous regions (tumors) based on the controlled overheating of target organs and tissues.

The biological premise of this treatment is the fact that tumor cells are more sensitive to high temperatures (specifically between 40°C and 45°C) than normal, healthy cells.

Hyperthermia is generally utilized as a highly effective adjuvant treatment. When combined with standard chemotherapy, radiotherapy, or both (known as Triple Modality), it allows for a reduction in anticancer drug doses, leading to fewer side effects for patients.

1.1. Physical Principles and Treatment Techniques

The basic technology behind oncological hyperthermia involves increasing tissue temperature via the absorption of electromagnetic fields at radio or microwave frequencies, which are radiated by dedicated external applicators (antennas). Because human tissue contains high water content, it acts as a dielectric material; the rapidly alternating electromagnetic fields cause water molecules to vibrate, generating heat through friction.

The primary physiological effects of this localized heating include:

- **Inhibition of repair mechanisms:** Heat stress prevents tumor cells from repairing the DNA damage caused by concurrent radiotherapy.
- **Vascular dilation:** Increased blood flow naturally facilitates a greater transport of chemotherapy agents directly to the tumor site.

Treatments are classified based on the tumor site (deep vs. superficial) and the tumor size (local, regional, or whole-body). Despite its benefits, hyperthermia poses several technical challenges:

- **Tracking:** Difficulties in reaching and maintaining HT temperatures at the desired target.
- **Collateral Damage:** There is a constant risk of burning or damaging surrounding normal cells if the specific absorption rate (SAR) is poorly distributed.
- **Timing & Standardization:** The treatment typically requires a longer duration and yields lower target accuracy compared to standard radiotherapy, and standardizing the procedure is difficult due to varying tumor locations, sizes, and morphologies.

2. The ALBA ON4000 System Setup

To address these challenges, we analyze the ALBA ON4000 system, which features compact, integrated technology designed for superficial hyperthermia. The traditional setup utilizes a passive arm to manually move an external applicator over the patient.

2.1. Technological Improvement through Tracking

To improve the manual setup and ensure accurate transfer of the pre-operative plan, an external optical tracking system was introduced.

- **Hardware:** An NDI Polaris Vega optical camera sensor coupled with a frame of spherical, highly reflective markers.

- **Function:** Markers are mounted directly on the antenna to provide real-time tracking.

Because we now have a physical camera, a tool (antenna), and pre-operative images (CT scans), multiple different reference frames are involved, making robust registration an absolute necessity.

3. Registration Problem Formulation

The ultimate goal of registration in this context is to register the pre-operative CT scan frame ($\mathcal{F}_{CT}$) against the visual tracking camera frame ($\mathcal{F}_V$).

Let us consider a target T with a known position in the CT scan, denoted as ${}^{CT}p_T$. To act upon this target with the applicator, we need to determine its location relative to the applicator frame ($\mathcal{F}_A$).

The transformation chain is formulated as:

$${}^Ap_T = {}^AT_{A_m} {}^{A_m}T_V {}^VT_{CT} {}^{CT}p_T \quad (1)$$

Where the transformations represent:

- ${}^AT_{A_m}$: The transformation from the applicator markers to the applicator body. This is a constant value available from the antenna CAD model or through an initial tool calibration.
- ${}^{A_m}T_V$: The transformation from the optical tracking camera to the markers. This is a time-varying value measured dynamically by the camera sensor.
- ${}^VT_{CT}$: The transformation from the CT scan to the visual frame. This is the constant output of the registration algorithm.

By solving for ${}^VT_{CT} = \arg \min d({}^VF, T_{CT} {}^{CT}F)$, the exact location of the target becomes fully known within the applicator frame $\mathcal{F}_A$.

4. Point-to-Point Registration (ICP Algorithm)

Note on Modern Approaches: While traditional Iterative Closest Point (ICP) algorithms remain widely used due to their robustness, state-of-the-art research is increasingly utilizing machine learning-based algorithms, especially for the initialization and point-matching phases of registration.

The core method utilized here is the point-to-point ICP algorithm, which relies on a matching between a set of N points ($N \geq 4$ for reliable 3D registration) in one coordinate frame and the corresponding N points in the other. Correspondences are often built using radiopaque anatomical landmarks placed on the patient's skin, which are visible in the pre-operative medical images (${}^{CT}p_i$) and measurable externally by the NDI camera (Vp_i).

4.1. Solving with Known Correspondences

If the point correspondences are explicitly known (i -th point in CT maps to i -th point in Vision), the ICP operation simplifies into a standard least-squares problem.

4.1.1. The Pseudo-Inverse Approach

The simplest analytical approach is to formulate the problem linearly and solve it via a pseudo-inverse. However, due to noisy point correspondences from sensor inaccuracies or skin movement, recovering a mathematically proper homogeneous transformation matrix (where the rotation matrix $R \in SO(3)$) is not guaranteed. The pseudo-inverse often yields a matrix that shears or

scales the space. Therefore, the pseudo-inverse solution is strictly used as an **initial guess** for iterative methods.

4.1.2. The Gauss-Newton Algorithm

To refine the initial pseudo-inverse guess and enforce the constraints of a valid rigid transformation, we utilize the non-linear Gauss-Newton optimization method.

We define our state vector as $x = (t_x, t_y, t_z, R(\alpha, \beta, \gamma)) \in SE(3)$, representing the 3D translation and the Euler angles of rotation. The objective is to minimize the sum of squared residual errors:

$$F(x) = \sum_{i=1}^N e_i^T(x) \Omega_i e_i(x) \quad (2)$$

where $e_i(x) = {}^V p_i - T(x)^{CT} p_i$ is the error vector for the i -th correspondence, and Ω_i is the inverse of the covariance matrix, which weighs the confidence in each measurement.

Because $F(x)$ is non-linear with respect to the angles in $SE(3)$, we linearize the error function around the current estimate x using a first-order Taylor expansion:

$$e_i(x + \Delta x) \approx e_i(x) + J_i \Delta x \quad (3)$$

where J_i is the Jacobian matrix of the error function with respect to the state x .

Substituting this approximation into our objective function yields a quadratic approximation of the cost:

$$F(x + \Delta x) \approx \Delta x^T H \Delta x + 2b^T \Delta x + c \quad (4)$$

Here, the terms are accumulated over all N points:

- The approximate Hessian matrix: $H = \sum J_i^T \Omega_i J_i$
- The gradient vector: $b = \sum J_i^T \Omega_i e_i(x)$
- The scalar cost: $c = \sum e_i^T(x) \Omega_i e_i(x)$

To find the optimal step Δx^* that minimizes this quadratic function, we take the derivative with respect to Δx and set it to zero, leading to the normal equations:

$$H \Delta x^* = -b \implies \Delta x^* = -H^{-1} b \quad (5)$$

The algorithm proceeds iteratively, updating the state $x \leftarrow x + \Delta x^*$ until the incremental correction Δx^* falls below a predefined convergence threshold.

4.2. Solving with Unknown Correspondences

If the correspondences between points are unknown, the ICP algorithm must solve the *data association* problem before acting as a least-squares solver. While KD-trees or Distance Maps are popular for dense point clouds, a **brute-force distance matrix** approach can be highly effective when utilizing a small number of artificial landmarks ($N \approx 4$).

4.2.1. Distance Matrix Method

The mathematical foundation of this method relies on the property of **isometry**: a rigid body transformation preserves the Euclidean distance between any two points. Therefore, the distance between marker 1 and marker 2 in the CT frame must perfectly equal the distance between marker 1 and marker 2 in the Camera frame.

Given an unsorted set of points ${}^V p$ and ${}^{CT} p$, we construct two symmetric distance matrices:

1. **Visual Distance Matrix (V_P):** Evaluate the relative distances between every pair of points in the camera frame. The element at row m , column n is $d_{m,n} = \|V p_m - V p_n\|$.
2. **CT Distance Matrix (CT_P):** Evaluate the relative distances between every pair of points in the CT frame. The element at row k , column l is $d_{k,l} = \|CT p_k - CT p_l\|$.

Because the points are unsorted, the matrices will not immediately match. However, the set of distance values contained in the rows/columns acts as a unique topological ‘signature’ for each point. By comparing the distance signatures in V_P against CT_P and finding the minimum differences, the algorithm can match row m in the visual frame to row k in the CT frame. Once sorted and matched, the algorithm proceeds to the Gauss-Newton optimization step.

5. Validation Architecture

To safely validate the registration algorithms, a custom software architecture was designed to interface the real NDI Polaris camera with the V-REP (CoppeliaSim) robotic simulator.

A realistic 3D model of a professional abdomen phantom, containing detailed inner tissues and organs, was reconstructed from CT scans using 3D Slicer and imported into the virtual scene. The customized software architecture interacts via three main classes:

1. **PolarisProxy class:** Queries the NDI Polaris camera using NDI C++ APIs to read the exact pose of the physical marker frames in the real world.
2. **RegistrationTask class:** Processes the unsorted data from the PolarisProxy and the pre-acquired CT models, runs the distance-matrix data association and Gauss-Newton ICP algorithm, and outputs the resulting transformations.
3. **VREPProxy class:** Updates the visual V-REP scene using the simulator’s remote APIs, applying the alignment data sent from the RegistrationTask.

Physical validation, performed via palpation of the physical phantom with the tracked probe, confirmed high system precision with a Root Mean Square (RMS) error of less than 1cm.

6. Summary and Key Conceptual Threads

- **Universality of the Registration Problem.** Even though the hyperthermia setup (applicator, visual camera, CT scan) utilizes very different hardware compared to orthopedic or neurosurgery systems, the fundamental need to align a digital pre-operative plan with a real-world interventional frame remains identical.
- **Technological Enhancements for Hyperthermia.** Introducing external optical tracking (NDI Polaris) transforms a purely manual therapy into a trackable, repeatable procedure, ensuring proper focus of radio/microwave frequencies to improve treatment efficacy.
- **Optimization over Analytical Solutions.** While point correspondences can be fed into a pseudo-inverse formulation, iterating through Gauss-Newton optimization is preferred to strictly enforce the properties of a valid rigid body transformation ($SO(3)$ constraint).
- **Data Association through Isometry.** When matching unknown sets of points, constructing distance matrices leverages the geometric fact that rigid transformations preserve Euclidean distances, creating unique ‘signatures’ to identify homologous points.
- **Simulation for Validation.** Custom architectures linking physical sensors (PolarisProxy) to rigid solvers (RegistrationTask) and visual digital twins (VREPProxy) are essential for safely verifying RMS errors before clinical deployment.

Key topics for revision.

1. The biological principles behind oncological hyperthermia and its advantages as an adjuvant treatment.
2. The kinematic chain of transformations required to locate a target T in the applicator frame $\mathcal{F}_A$.
3. Solving the ICP algorithm with known correspondences: pseudo-inverse vs. Gauss-Newton algorithm (linearization, Jacobian, Hessian, and normal equations).
4. Solving the ICP algorithm with unknown correspondences: the brute-force distance matrix method and why it works.